Infraestrutura,Incubação,Startup.
Projectos devo entregar.
Objectives Machine learning.
Artificial Neural Network, for machine learning
Feed Forward Neural Network, for prediction
Convolutional Neural Network, for Image Analyses
Latent Space and Generative AI,LLM

Prerequisites.
Python basics
Data management and Visualization tools (basics) – Pandas,Matplotlib. //To download
Python crash course
https://tinyurl.com/y8z83z9k
SCIKIT_LEARN for machine learning
Tensor flow or PyTorch(better) for Artifical Neural Network.

Machine Learning
Supervised is when I and my data teach the AI.
Unsupervised is when the machine tries to found the pattern alone. Reinforced Learning. Where we give rewards for a specific pattern to have the best reward.
We are going to focus in **Supervised learning** and **Artificial Neural Network**.
Describe the environment, the actions and the actions.

Ex: If you have many houses, each described by its size,distance from the city center,year of construction, can you teach a computer to estimate the price of a new house?
The computer will predict a number (0,+infinity).
Input house data
Input → Model → Cost, this is **regression**

Ex2: If you have many images, can you teach a computer to distinguish pictures of cats and pictures of dogs?
Input image
The computer will give a probability of beign a dog or cat. It predicts the class
Classification

Ex3: If you have seen many images of cats, can you teach a computer to generate the image of a cat?
Input Latent Space/Random code
Generative AI
To generate an image first we give some samples to the computer, then the computer will search for patterns to generate other images.
Then we give something a bit different.
For example supose we have an image of 200x200 pixels, we have 2**40000 possibilities //TODO

How to do (without ML)?
For Ex1:
if #BR > 4
  if size < 200m²
   if age < 40
    cost = 2000

We create a rule for each condition with if-elif-else.
But in ML.
Instead of programming for each case,we select a model to learn(train) with our datas and then with the new model, we give a data that it never saw and it will make a new prediction, we can use the school as an analogy.

Definition that describe ML
A **system(model)** learns from experience E
with respect to a **task** T (Regression)
and a **performance measure** P (real - pred)
if its performance on T,
as measured by P,
improves with E.

How does experience look like?
Experience is the preview data, in our case experience is a table.
The atributes are **Features** and it will be the input, because we will give the model new Features that we don’t the price.
What we want to predict is the **Target(Label)** and it won’t be in the input,
To avoid the overfitting we have.
Inductive Bias (we choose what type of function the model will use)
Regularization
In reality LLM under the hood are autoregressive.

To avoid error we use
NO FREE LUNCH THEOREM
That states there isn’t models that always works in every situations.
To make the best model we should give it diverse inputs

https://tinyurl.com/22tszy4s